

RaceDay — API Endpoint Plan

Student Name	Sade Sazila
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Introduction

This document lists every planned RESTful API endpoint for the RaceDay event management system. Each endpoint specifies the HTTP method, the route, a short description, the role required to access it, the expected request body, and the expected response. All endpoints follow REST conventions and use session-based authentication.

1. Authentication Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Register a new user account	Public	{ firstName, lastName, email, password, role, phoneNumber, dateOfBirth }	201 Created — user object; 400 Bad Request — validation errors
POST	/api/auth/login	Login with credentials and create a session	Public	{ email, password }	200 OK — session created; 401 Unauthorized — invalid credentials
POST	/api/auth/logout	End the current authenticated session	Any Logged In	None	204 No Content — session cleared

2. User Profile Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/profile	Get current logged-in user profile	Any Logged In	None	200 OK — user profile object; 401 Unauthorized
PUT	/api/users/profile	Update current user's profile details	Any Logged In	{ bio, city, phoneNumber }	200 OK — updated profile; 400 Bad Request
PUT	/api/users/profile/picture	Upload a profile picture	Any Logged In	{ imageFile }	200 OK — image URL; 400 Bad Request

3. Events Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Get all upcoming events	Public	None	200 OK — list of events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{id}	Get a specific event by ID	Public	None	200 OK — event object; 404 Not Found
POST	/api/events	Create a new event	Organizer	{ name, description, eventDate, location, distance, eventType, imageURL }	201 Created — event object; 400 Bad Request; 403 Forbidden
PUT	/api/events/{id}	Update an existing event	Organizer	{ name, description, eventDate, location, distance, eventType, imageURL }	200 OK — updated event; 403 Forbidden; 404 Not Found
DELETE	/api/events/{id}	Delete an event and cascade related data	Organizer	None	204 No Content; 403 Forbidden; 404 Not Found
PUT	/api/events/{id}/banner	Upload an event banner image	Organizer	{ imageFile }	200 OK — image URL; 403 Forbidden

4. Categories Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{id}/categories	Get all categories for a specific event	Public	None	200 OK — list of categories; 404 Not Found
GET	/api/categories/{id}	Get a single category by ID	Public	None	200 OK — category object; 404 Not Found
POST	/api/events/{id}/categories	Add a new category to an event	Organizer	{ name, minAge, maxAge }	201 Created — category object; 403 Forbidden
PUT	/api/categories/{id}	Update an existing category	Organizer	{ name, minAge, maxAge }	200 OK — updated category; 403 Forbidden; 404 Not Found
DELETE	/api/categories/{id}	Delete a category from an event	Organizer	None	204 No Content; 403 Forbidden; 404 Not Found

5. Enrollments Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/enrollments/my	Get all enrollments for the logged-in participant	Participant	None	200 OK — list of enrollments; 401 Unauthorized
GET	/api/enrollments/{id}	Get a single enrollment by ID	Any Logged In	None	200 OK — enrollment object; 403 Forbidden; 404 Not Found
GET	/api/events/{id}/enrollments	Get all enrollments for a specific event	Organizer	None	200 OK — list of enrollments; 403 Forbidden
POST	/api/enrollments	Enroll a participant in an event with a category	Participant	{ eventId, categoryId }	201 Created — enrollment object; 400 Bad Request; 409 Conflict — already enrolled
DELETE	/api/enrollments/{id}	Cancel an enrollment	Participant	None	204 No Content; 403 Forbidden; 404 Not Found

6. Results Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/results/my	Get all results for the logged-in participant	Participant	None	200 OK — list of results; 401 Unauthorized
GET	/api/results/{id}	Get a single result by ID	Any Logged In	None	200 OK — result object; 403 Forbidden; 404 Not Found
GET	/api/events/{id}/results	Get all results for a specific event	Organizer	None	200 OK — list of results; 403 Forbidden
POST	/api/results	Capture a result for an enrollment	Organizer	{ enrollmentID, finishTime, finishPosition, totalFinishers }	201 Created — result object; 403 Forbidden; 404 Not Found
PUT	/api/results/{id}	Update an existing result	Organizer	{ finishTime, finishPosition, totalFinishers }	200 OK — updated result; 403 Forbidden; 404 Not Found

7. HTTP Status Code Reference

Code	Meaning
200 OK	Successful retrieval or update
201 Created	Successful creation of a new resource
204 No Content	Successful deletion or logout
400 Bad Request	Validation error in the request
401 Unauthorized	Not authenticated
403 Forbidden	Authenticated but wrong role
404 Not Found	Resource does not exist
409 Conflict	Duplicate record (e.g. already enrolled)

References

Microsoft Corporation. 2024. ASP.NET Core Web API documentation. [Online]. Available at: <https://learn.microsoft.com/en-us/aspnet/core/web-api/> [Accessed: 16 September 2026].

Richardson, L. and Amundsen, M. 2013. RESTful Web APIs. O'Reilly Media.